

Disulfide Connectivity Prediction with Normalized Bond Vector

Lin-Yu Tseng¹, Shiao-Kuen Tsai¹ and Hsuan-Hung Lin^{2,3}

¹Department of Computer Science, National Chung Hsing University,
250 Kuo Kuang Road, Taichung, Taiwan
lytseng@cs.nchu.edu.tw

²Department of Applied Mathematics, National Chung Hsing University,
250 Kuo Kuang Road, Taichung, Taiwan

³Department of Management Information Systems, Central Taiwan University of Science
and Technology, Taichung 40601 Taiwan

Abstract

The prediction of the location of disulfide bridges helps towards the solution of protein folding problem. Most of previous works on disulfide connectivity pattern prediction use the prior knowledge of the bonding state of cysteines. In this study an effective method is proposed to predict disulfide connectivity pattern for proteins that have at least two disulfide bonds but without the prior knowledge of cysteines' bonding state. The method is based on template matching and only a global feature called the normalized bond vector is used. This method can also be applied and achieves a better accuracy if the bonding state of cysteines is known. In previous research works reported in the literature, to the best of our knowledge, without the prior knowledge of the bonding state of cysteines, the best accuracy of disulfide connectivity pattern prediction is 51%. With the prior knowledge of the bonding state of cysteines, the best accuracy is 70%. Testing our method on the same test dataset SwissProt39 (sp39), the accuracy of a fourfold cross-validation is 61% for disulfide connectivity pattern prediction when the bonding state of cysteines is not known in advance. With the prior knowledge of bonding state, the accuracy increases to 71%.

1 Introduction

Disulfide bonds play an import structural role in stabilizing protein conformations. The prediction of disulfide bonding pattern helps to a certain degree prediction of the three-dimensional protein structure and hence its function because disulfide bonds impose geometrical constraints on the protein backbones. Some recent research works do have shown the close relation between the disulfide bonding patterns and the protein structures [7][15].

In the realm of the disulfide bond prediction,

two problems are addressed. The first is the prediction of the disulfide bonding states and the second is the prediction of the disulfide bonding pattern. Recently, significant progress has been made in the prediction of the disulfide bonding states. Several methods based on statistical analysis [12], neural networks [10][13], or support vector machines [5] have been proposed and they are quite effective in predicting the bonding state of cysteines with the accuracy around 81%-90%.

Recently, there are also several methods that were proposed for the prediction of the disulfide bonding pattern. The first method was presented by Fariselli and Casadio [8]. They reduced disulfide connectivity to the graph matching problem in which vertices are oxidized cysteines and edges are labeled by the strength of interaction (contact potential) in the associated pair of cysteines. The Monte Carlo simulated annealing method is used to find the optimal values of contact potentials and finally the disulfide bridges are located by finding the maximum weight perfect matching. Fariselli et al. [9] improved their results by using neural networks to predict the cysteine pairwise interactions. Vullo and Frasconi [16] developed an ad-hoc recursive neural network for scoring labeled undirected graphs that represent connectivity pattern and they improved the accuracy of bonding pattern prediction significantly from 34% to 44%. Baldi *et al.* [2] improved the accuracy by using two-dimensional recursive neural networks to predict bonding probabilities between cysteine pairs. Ferrè and Clote [11] used the diresidue neural network to predict bonding probabilities between cysteine pairs. They also used secondary structure information and diresidue frequencies in their training. Tsai *et al.* [14] used the support vector machine to predict bonding probabilities between cysteine pairs. The features used in training the support vector machine are local sequence profiles and the linear distance of cysteines. All above mentioned methods are based on the reduction of the bonding pattern prediction to the maximum

weight perfect matching problem. The following three methods are not based on this reduction. Chen and Hwang [4] used the support vector machine to predict the bonding pattern directly. The features they used in training the support vector machine are the coupling between the local sequence environments of cysteine pairs, the cysteine separations, and the amino acid content. Zhao *et al.* [17] used a simple feature called cysteine separations profiles (CSP) to predict the bonding patterns. Chen *et al.* [3] used a two-level model and achieved the accuracy of 70% that is the best accuracy for the present. All these methods except two [2][6] assume that the bonding states are known. The method proposed by Baldi *et al.* [2] can be applied whether the bonding states are known or not.

In this work, a template matching method to predict the bonding pattern of cysteines is presented. Only one global feature called the normalized bond vector (NBV) is used. Knowing in advance that the proteins have only 2 to 5 disulfide bonds and without the prior knowledge of bonding states, the accuracy of bonding pattern prediction is 61%. With the prior knowledge of bonding states, the accuracy is 71%.

2 System And Methods

2.1 The Dataset

In order to compare our method with previously reported methods [16][2][14], the same dataset extracted from the SWISS-PROT database release no. 39 (SP39) [1] was employed. This set was obtained by applying the same filtering procedure [8] in order to include only high quality and experimentally verified intra-chain disulfide bridge annotations. It is noted that less than 20% of SWISS-PROT sequences have more than five disulfide bridges. In this set, all the chains having more than 10 oxidized cysteines had been excluded. There are in total 446 sequences, of which 156 sequences have two disulfide bridges, 146 sequences have three, 99 sequences have four, and 45 sequences have five. For the four-fold cross-validation procedures, this dataset was further randomly divided into four sets. In such a way that the number of sequences of each set is approximately equal.

Table 1. Number of possible disulfide bonding patterns (N_p) and number of actual disulfide bonding patterns (N_a) for protein chains with different disulfide bridge numbers

Number of disulfide bridges (B)	N_p^a	N_a^b
B=2	3	3
B=3	15	15
B=4	105	43
B=5	945	45
B=6	10395	29
B=7	135135	14

^aNumber of possible disulfide bonding patterns.

^bNumber of actual disulfide bonding patterns.

Statistics obtained from the SSDB database (<http://e106.life.nctu.edu.tw/~ssbond/>) [7]

2.2 The number of actual disulfide bonding patterns

The number of possible disulfide bonding patterns increases exponentially with the number of disulfide bridges. The number of possible disulfide bonding patterns of a protein with n disulfide bridges, denoted by N_p , can be calculated as follows:

$$N_p = \frac{\binom{2n}{2} \binom{2n-2}{2} \dots \binom{2}{2}}{n!} = (2n-1)!! = \prod_{i \leq n} (2i-1)$$

Table 1 lists the number of possible disulfide bonding patterns and the number of actual disulfide bonding patterns with disulfide bridge number ranging from 2 to 7. The number of actual disulfide bonding patterns denoted by N_a is obtained from SSDB database [7]. It is noted in Table 1 that although the number of possible disulfide bonding patterns increases rapidly, the number of actual disulfide bonding patterns has a peak value of 45 at $B = 5$ and does increase mildly. This fact makes the template matching of our method possible.

2.3 The normalized bond vector (NBV)

A global feature is used to evaluate the similarity between the test protein sequence and a template protein sequence in our template matching method. We predict the disulfide bonding pattern of the test protein sequence to be the same as that of the most similar template sequence. This feature is called the normalized bond vector (NBV). It is described in the following and illustrated by an example shown in Figure 1.

The protein 1BET has six cysteines and three disulfide bridges are formed with the six cysteines.

The positions of these six cysteines and the disulfide bonding pattern [1-4, 2-5, 3-6] are shown

in Figure 1(a).

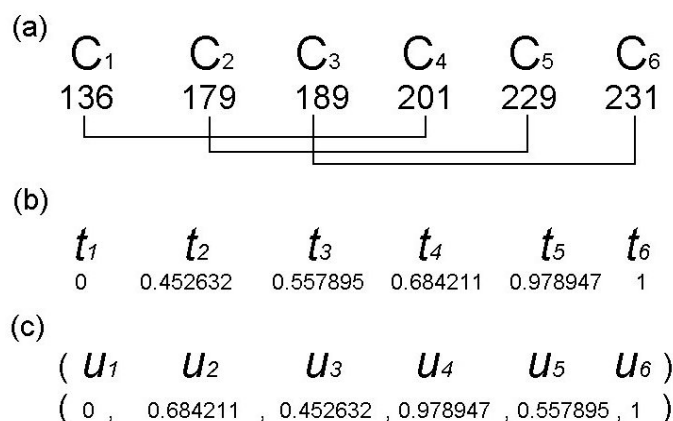


Figure. 1. The NBV of protein 1BET: (a) the positions of six cysteines and the bonding pattern, (b) the normalized positions of six cysteines, (c) the normalized bond vector.

Let the position of each cysteine C_i be denoted by s_i . The normalized position t_i of each cysteine C_i is computed as follows.

FOR B > 2

$$t_i = \frac{s_i - s_1}{s_{2B} - s_1}, \quad i = 1, 2, \dots, 2B$$

FOR B = 2

$$t_i = \frac{s_i - \bar{s}}{s_4 - s_1}, \quad i = 1, 2, 3, 4$$

Where $\bar{s} = \frac{1}{4} \sum_{i=1}^4 s_i$

The formula for B = 2 is a little bit different from others because the discrimination ability is better using $s_i - \bar{s}$ instead of $s_i - s_1$. Figure 1(b) shows the normalized positions of six cysteines. Having the normalized position of each cysteine, the normalized bond vector (NBV) is defined as a vector $(u_1, u_2, \dots, u_{2B})$ of 2B components. Each component of NBV is a normalized position t_i and the order of t_i 's coincides with that of the bonding pattern. As an example, Figure 1(c) shows the NBV of the protein 1BET.

2.4 The template matching

Our method is based on the template matching. The NBV of each protein sequence in the training set is computed and used as a template. All templates are divided into groups according to the number of disulfide bridges. So there are four sets of templates in our experiment on the dataset SP39, each for B = 2, 3, 4 and 5.

As mentioned before, the number of the actual disulfide bonding patterns is not large. This makes the template matching possible. When a test

protein sequence is given, since we do not know the oxidation states of cysteines, we do not know the number of disulfide bridges either. In our method, all possible oxidation states of cysteines are enumerated and for each possible oxidation state of cysteines, all possible templates are matched. For example, given a test protein with seven cysteines, all C(7,6) possible oxidation states are enumerated and each one is matched with all templates that have three disulfide bridges. After that, all C(7,4) possible oxidation states are enumerated and each one is matched with all templates that have two disulfide bridges. At the first glance, the number of template matches may seem huge. Assuming that there is exactly one template for each bonding pattern, we compute the number of template matches for a test protein with ten cysteines and another test protein with fifteen cysteines in the following to show that this is not the case. The number of matches for a test protein with ten cysteines is 5,760 as computed below.

$$\binom{10}{10} * 45 + \binom{10}{8} * 43 + \binom{10}{6} * 15 + \binom{10}{4} * 3 = 5760$$

While the number of matches for a test protein with fifteen cysteines is 491,010 as computed below.

$$\binom{15}{10} * 45 + \binom{15}{8} * 43 + \binom{15}{6} * 15 + \binom{15}{4} * 3 = 491010$$

In testing our method on dataset SP39, since there may be several templates for each bonding pattern, the numbers of matches for a test protein with ten cysteines and a test protein with fifteen cysteines are estimated to be 50,824 and 1,283,543, which can be managed by a computer easily.

The distance D between the test protein sequence and a template that has B disulfide bridges is defined in the following.

$$D = \frac{1}{B} \left(\sum_{i=1}^B \sqrt{(u_{2i-1}^{(1)} - u_{2i-1}^{(2)})^2 + (u_{2i}^{(1)} - u_{2i}^{(2)})^2} \right)$$

Where $(u_1^{(1)}, u_2^{(1)}, \dots, u_{2B}^{(1)})$ and $(u_1^{(2)}, u_2^{(2)}, \dots, u_{2B}^{(2)})$ are the NBV of the test protein sequence and the template respectively. An example is stated below and illustrated in Figure 2. The protein 1B8M has seven cysteines and three disulfide bridges with

bonding pattern [2-5, 3-6, 4-7] (shown in Figure 2(a)). C_2, C_3, C_4, C_5, C_6 and C_7 are oxidized cysteines. The NBV corresponding to this sequence is listed in Figure 2(b). Then, the distance between the test protein 1B8M and the template protein 1BET (shown in Figure 1.) is calculated as follows.

$$D = \frac{1}{3} (\sqrt{(0-0)^2 + (0.701923-0.68411)^2} + \sqrt{(0.423077-0.452632)^2 + (0.980769-0.978947)^2} + \sqrt{(0.586538-0.557895)^2 + (1-1)^2}) = 0.025322322$$

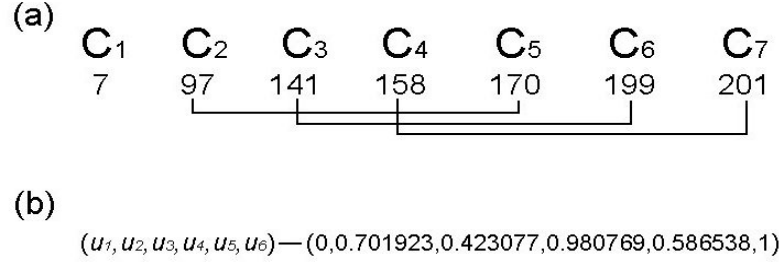


Figure 2. The NBV of protein 1B8M: (a) the positions of cysteines in 1B8M and the bonding pattern, (b) NBV of 1B8M with oxidized cysteines

```

If  $\left\lfloor \frac{n}{2} \right\rfloor \leq 5$  then  $K = \left\lfloor \frac{n}{2} \right\rfloor$  else  $K = 5$ 
flag = 0 ; threshold5 = 0.035 ; threshold4 = 0.035 ; threshold3 = 0.030 ; threshold2 = 0.025
For i = K down to 2
    Enumerate all C (n, 2K) possible subsets of cysteines
    For each subset of cysteines
        Compute the NBV of the test protein sequence.
        Compute the distance D of this NBV and each template in the template set with B = i.
        Choose the template with the smallest D (let this value be denoted by D0)
        in the template set with B = i and store its bonding pattern in ANS.
    End For
    If flag = 0 then
        ANSℓ ← ANS ; Dℓ ← D0 ; flag ← 1
    else if (D0 ≥ 0.04) or (Dℓ - D0 ≤ thresholdi) then
        output ANSℓ and STOP
    else if i > 2 then
        ANSℓ ← ANS ; Dℓ ← D0
        else output ANS and STOP
    End For
End For
    
```

Figure 3. The algorithm for disulfide bonding pattern prediction.

3 Algorithm

The algorithm for disulfide bonding pattern prediction knowing in advance that the proteins have only 2 to 5 disulfide bonds and without the prior knowledge of the oxidation states of cysteines is described in the following.

The NBV of each protein sequence in the training set is computed and taken as a template. Four template sets are constructed, each with $B = 2, 3, 4$, and 5 . Whenever a test protein sequence with n cysteines is given, first it will be matched with the templates in the template set with $B = \lfloor \frac{n}{2} \rfloor$ ($B = 5$ if $\lfloor \frac{n}{2} \rfloor > 5$) and along down to the template set with $B=2$. For example, a test protein sequence with seven cysteines is first matched with the template set that has three disulfide bridges. After that, it is matched with the template set that has two disulfide bridges. The details of the template matching algorithm are presented in Figure 3.

An example is presented below to illustrate the algorithm. Suppose the protein 1TPA is given as a test protein, since 1TPA has eight cysteines in its sequence, it is first matched with all the templates in the template set with $B = 4$. In this template set, the template most similar to 1TPA is 1GPT with the distance $D = 0.0356788$ and the bonding pattern [1-8, 2-5, 3-6, 4-7]. Then, all $C(8,6)$ subsets of six cysteines are enumerated and for each subset, it is matched with all the templates in the template set with $B = 3$. After all the enumerations and the matchings with templates, the subset $\{C_3, C_4, C_5, C_6, C_7, C_8\}$ is found to be most similar to the template protein 1AAP with the distance $D = 0$ and the bonding pattern [1-6, 2-4, 3-5]. Since $D=0$ is less than 0.04 and $0.0356788 - 0 = 0.0356788 > \text{threshold3} = 0.030$, the algorithm continues to enumerate all $C(8,4)$ subsets of four cysteines and to match each subset with all the templates in the template set with $B = 2$. After all the enumerations and the matchings with templates, the subset $\{C_3, C_4, C_5, C_8\}$ is found to be most similar to the template protein 1BRV with the distance $D = 0.003879$ and the bonding pattern [1-4, 2-3]. Since $0 - 0.003879 < \text{threshold2} = 0.025$, the algorithm will output $\{C_3, C_4, C_5, C_6, C_7, C_8\}$ with the bonding pattern [1-6, 2-4, 3-5] (i.e. the bonding pattern $[C_3-C_8, C_4-C_6, C_5-C_7]$) as its prediction result. This prediction is indeed correct.

The algorithm can also be applied if the oxidation states of cysteines in the test protein sequence are known in advance. If there are $2K$ oxidized cysteines, the test protein sequence needs only to be

matched with all the templates in the template set with $B = K$ and the bonding pattern of the template that has a smallest value of D will be output as the answer.

4 Results and Discussion

In order to compare the proposed method with previous approaches [16][2][14], the same dataset SP39 was used. For four-fold cross-validation, the dataset SP39 was randomly divided into four sets, such that each set contains approximately equal number of sequences.

Table 2 and Table 3 show the results of our method and the comparison with previous approaches reported in the literature. Table 2 lists prediction accuracies obtained without the prior knowledge of the oxidation state of cysteines. Table 3 lists prediction accuracies with the prior knowledge of the oxidation state of cysteines.

Q_P is defined as follows:

$$Q_P = \frac{C_P}{T_P}$$

,

where C_P is the number of proteins whose bonding patterns are correctly predicted and T_P is the total number of proteins in the test set.

To the best of our knowledge, only the methods proposed by Baldi *et al.* [2] and Cheng *et al.* [6] can be applied without the prior knowledge of the oxidation states of cysteines. Testing on the same dataset SP39, the accuracy of our method is very competitive as shown in Table 2. As for the case when the oxidation states of cysteines are known in advance, the accuracies achieved by our method are also significantly better than those of other methods. It is noted that the accuracies for $B = 3$ and 4 are much higher than those of other methods. This reveals that the global feature NBV has a good discrimination ability for $B = 2, 3$ and 4 (the accuracies range from 71% to 78%). The accuracies for $B = 5$ is significantly lower than those of the method proposed by Chen *et al.* [3]. This shows that using some local sequence profiles as the feature may further improve the accuracies. It is also noted that with the prior knowledge of the oxidation states of cysteines, the accuracies for $B = 2$ and 3 have greatly improved over those without this knowledge. This is because many subsets of four cysteines and subsets of six cysteines have similar NBV and therefore the prediction is prone to error.

Table 2. Prediction accuracies without the prior knowledge of the oxidation states of cysteines

Methods	B = 2 $Q_P(\%)$	B = 3 $Q_P(\%)$	B = 4 $Q_P(\%)$	B = 5 $Q_P(\%)$	B = 2-5 $Q_P(\%)$
2D-Rnn profile ^a	40	32	15	3	28
NBV template matching	65	54	75	38	61
	B = 1	B = 2	B = 3	B = 4	Overall(1-26)
2D-Rnn profile ^b	59	59	54	34	51

^aBaldi *et al.* (2005)^bCheng *et al.* (2006)

Table 3. Prediction accuracies with the prior knowledge of the oxidation states of cysteines

Methods	B = 2 $Q_P(\%)$	B = 3 $Q_P(\%)$	B = 4 $Q_P(\%)$	B = 5 $Q_P(\%)$	B = 2-5 $Q_P(\%)$
MC graph-matching ^a	56	21	17	2	29
NN graph-matching ^b	68	22	20	2	34
BiRnn-2 profile ^c	73	41	24	13	44
2D-Rnn profile ^d	74	51	27	11	49
dNN2 ^e	62	40	55	26	49
SVM ^f	74	61	30	12	55
Pair-wise SVM ^g	79	53	55	58	63
2-level SVM ^h	85	67	57	58	70
NBV template	78	71	75	38	71

^aFrisiselli and Casadio (2001).^bFrisiselli *et al.* (2002).^cVuillo and Frasconi (2004).^dBaldi *et al.* (2005).^eFerrè and Clote (2005). Only results of Q_P are available.^fChen and Hwang (2005).^gTsai *et al.* (2005).^hChen *et al.* (2006)

5 Conclusion

Recently, the progress in the prediction of the oxidation states of cysteines in protein sequence is significant. But for the prediction of the bonding pattern of cysteines, much research effort is still needed to improve the prediction accuracies. To these authors' knowledge, all previous approaches except the two presented by Baldi *et al.* [2] and Cheng *et al.* [6] assume that the oxidation states of cysteines are known in advance. To practically solve the prediction of the bonding pattern of cysteines, this assumption eventually should be removed. In this work, without the prior knowledge of the oxidation states of cysteines, a template matching method based on a global feature called the normalized bond vector (NBV) achieves the accuracy of 61% in the bonding pattern prediction, which improves the previous accuracy of 51% obtained by Cheng *et al.* [6]. If the oxidation states of cysteines are known in advance, our method can also be applied to predict the bonding pattern of cysteines. It achieves the accuracy of 71%, which is also better than the best accuracy of 70% reported previously [3]. This

study reveals the importance of the global feature in the prediction of the bonding pattern of cysteines.

The future work is to find a combination of global features and the local features that can further improve the accuracies of the prediction.

Acknowledgments. We would like to thank National Science Council of Taiwan, ROC for financial support under the contract number NSC94-2213-E-005-020. We would also like to thank Mr. Jianlin Cheng and Professor Pierre Baldi for providing us the test dataset SP39.

References

- [1] Bairoch, A., Apweiler, R.: The Swiss-Prot protein sequence database and its supplement TrEMBL in 2000. *Nucleic Acids Res.* 28 (2000) 45–48
- [2] Baldi, P., Cheng, J., Vullo, A. Large-scale prediction of disulphide bond connectivity. In: Saul, L. K., Weiss, Y., Bottou, L. (eds.): *Advances in Neural Information Processing Systems* 17. MIT Press, Cambridge, MA

- (2005) 97–104
- [3] Chen, B. -J., Tsai, C. -H. , Chan, C. -H., Kao, C. -Y.: Disulfide connectivity prediction with 70% accuracy using two-level models. *Proteins* 55(1) (2006) 246–252.
 - [4] Chen, Y.-C., Hwang, J. -K.: Prediction of disulfide connectivity from protein sequences. *Proteins* 61 (2005) 507–512
 - [5] Chen, Y.-C., Lin, Y. -S., Lin, C. -J., Hwang, J. -K.: Prediction of the bonding states of cysteines using the support vector machines based on multiple feature vectors and cysteine state sequences. *Proteins* 55 (2004) 1036–1042
 - [6] Cheng, J., Saigo, H., Baldi, P.: Large-scale prediction of disulphide bridges using kernel methods, Two-dimensional recursive neural networks, and weighted graph matching. *Proteins* 62 (2006) 617–629
 - [7] Chuang, C. -C., Chen, C. -Y., Yang, J. -M., Lyu, P. -C., Hwang, J. -K.: Relationship between protein structures and disulfide-bonding patterns. *Proteins* 55 (2003) 1–5
 - [8] Fariselli, P., Casadio, R.: Prediction of disulfide connectivity in proteins. *Bioinformatics* 17 (2001) 957–964
 - [9] Fariselli, P., Martelli, P. L., and Casadio, R.: A neural network base method for predicting the disulfide connectivity in proteins. In: Damiani, E. et al., (eds.) *Knowledge based Intelligent Information Engineering Systems and Allied Technologies KES 2002*, IOS Press, Amsterdam 1 (2002) 464–468
 - [10] Fariselli, P., Riccobelli, P., Casadio, R.: Role of evolutionary information in predicting the disulfide-bonding state of cysteine in proteins. *Proteins* 36 (1999) 340–346
 - [11] Ferrè, F., Clote, P.: Disulfide connectivity prediction using secondary structure information and diresidue frequencies. *Bioinformatics* 21 (2005) 2336–2346
 - [12] Fiser, A., Simon, I.: Predicting the oxidation state of cysteines by multiple sequence alignment. *Bioinformatics* 16 (2000) 251–256.
 - [13] Martelli, P. L., Fariselli, P., Malaguti, L., Casadio, R.: Prediction of the disulfide-bonding state of cysteines in proteins with hidden neural networks. *Protein Engineering* 15 (2002) 951–953
 - [14] Tsai, C. -H., Chen, B. -J., Chan, H.-H., Liu, H. -L., Kao, C. -Y.: Improving disulfide connectivity prediction with sequential distance between oxidized cysteines. *Bioinformatics* 21 (2005) 4416–4419
 - [15] van Vlijmen, H. W. T., Gupta, A., Narasimhan, L. S., Singh, J.: A novel database of disulfide patterns and its application to the discovery of distantly related homologs. *J. Mol. Biol.* 335 (2004) 1083–1092
 - [16] Vullo, A., Frasconi, P.: Disulfide connectivity prediction using recursive neural networks and evolutionary information. *Bioinformatics* 20 (2004) 653–659
 - [17] Zhao, E., Liu, H. -L., Tsai, C. -H., Tsai, H. -K., Chan, C. -H., Kao, C. -Y.: Cysteine separations profiles on protein sequences infer disulfide connectivity. *Bioinformatics* 21 (2005) 1415–1420